

CONNECT THE DOTS: NAVIGATING NETWORKS WITH BFS & DFS

Master the art of graph traversal through Indian metro maps and social circles.

| 1. The Social Connection

Have you ever wondered how you are connected to a celebrity through just a few friends? [cite: 4] This is because we all live in a giant **Social Network**—a web of connections where everyone you know is linked to someone else. [cite: 4]

Did you know?

Every person on Earth is roughly six "steps" away from any other person. In graph theory, these steps are the edges connecting the nodes (people). [cite: 4]

This web of friendships, family trees, and even city maps can be simplified into a structure called a **Graph**. [cite: 4]

| 2. Building Blocks: Nodes & Edges

A graph is not a chart with bars or lines—it is a collection of dots connected by paths. [cite: 4]

Nodes (Vertices)

The "dots" or points in a graph.
These represent entities like:

- Metro Stations [cite: 4]
- People in a circle [cite: 4]
- Cities on a map [cite: 4]

Edges

The "lines" that connect the dots.
These represent relationships like:

- Train tracks [cite: 4]
- Friendships [cite: 4]
- Flight paths [cite: 4]

Formula: Node = Point | Edge = Connection [cite: 4]

| 3. Real-World Case: Delhi Metro

The Delhi Metro is a perfect example of a graph in real life! [cite: 4]

- **Nodes:** Major interchange stations like **Rajiv Chowk** and **Hauz Khas**. [cite: 4]
- **Edges:** The actual tracks that connect these stations. [cite: 4]
- **The Network:** Every time you travel from one station to another, you are moving along edges between nodes. [cite: 4]

In this session, we will learn **how** a computer navigates this map to find the best route. [cite: 4]

| 4. Navigational Strategies

Once we have a graph, we need a strategy to explore it. In computer science, we call this **Traversal**. [cite: 4] We will follow two travelers with very different styles: **Buddy BFS** and **Daring DFS**. [cite: 4]

Buddy BFS

The "Nearby" Explorer. Buddy likes to stay close and check everything around him first. [cite: 4]

Daring DFS

The "Path" Explorer. Daring likes to pick one direction and go as deep as possible. [cite: 4]

5. Breadth-First Search (BFS)

Think of BFS like a **ripple in a pond** or a smell spreading through a house. [cite: 4] It moves layer by layer. [cite: 4]

The BFS Strategy:

1. Start at a node. [cite: 4]
2. Visit all **immediate neighbors** (1 step away). [cite: 4]
3. Only after visiting all of them, move to the neighbors 2 steps away. [cite: 4]

Best for: Finding the **Shortest Path**. If you want the station closest to you, Buddy BFS will find it first! [cite: 4]

6. Depth-First Search (DFS)

Think of DFS like exploring a **maze** or a multi-story building. [cite: 4] It picks one branch and follows it to the very end. [cite: 4]

The DFS Strategy:

1. Pick a path. [cite: 4]
2. Go as deep as possible until you hit a **dead end**. [cite: 4]
3. Backtrack to the last junction and try another deep path. [cite: 4]

Best for: Thorough exploration and visiting every corner of a large network. [cite: 4]

| 7. Scenario: The Metro Rescue

Imagine you lost your bag at a metro station. [cite: 4]

BFS Approach: You check every station that is just 1 stop away. Then all stations 2 stops away. [cite: 4]

Result: You find it quickly if it's nearby! [cite: 4]

DFS Approach: You pick one metro line and follow it all the way to the last station. [cite: 4]

Result: You might waste time going far away if the bag was just one stop behind you! [cite: 4]

| 8. Summary & Key Takeaways

- **Graphs:** Made of Nodes (dots) and Edges (lines). [cite: 4]
- **BFS:** Searches in **Layers**. Best for finding the nearest target fast. [cite: 4]
- **DFS:** Searches one **Path** fully. Best for exploring every corner of a maze. [cite: 4]
- **Logic:** Working together keeps information safe and networks navigable! [cite: 4]

Congratulations!

You've mastered the basics of Graph Theory. [cite: 4]